Power Electronics for Renewable Energy Systems

EE661 : Course Project

Average Current Mode Control of Buck Converter

Report submitted by-
Kashib khan (254102505)
Saurav Kr Yadav (25410112)

Under the supervision of-
Dr.Shabari Nath

Electronics and Electrical Engineering Department
Indian Institute of Technology Guwahati - 781039

Introduction

This chapter presents the analysis of an average current-mode controlled pulse-width modulated (PWM) buck DC–DC converter operating in continuous conduction mode (CCM). The modelling approach is based on circuit averaging using energy conservation. A small-signal model is developed considering practical losses. Open-loop and closed-loop transfer functions and transient responses are analysed. A design example is included to illustrate system behaviour.

Objectives

- Explain DC characteristics of the buck converter
- Develop small-signal linear model
- Analyze open-loop transfer functions
- Evaluate input and output impedances
- Design control circuit
- Study average current-mode control behavior
- Analyze inner current loop transfer functions
- Analyze outer voltage loop transfer functions

Assumptions

- Diode capacitance and stray inductance neglected
- Passive components are linear and time-invariant
- Input source has zero output impedance
- Converter operates in steady state
- Switching frequency is much higher than system time constants
- Output voltage is constant; input voltage and load may vary

1 Circuit Description, DC Characteristics, and Design

1.1 Circuit Description

The PWM buck converter circuit is illustrated in Figure 2.1. This converter operates in Continuous Conduction Mode (CCM). The circuit comprises a power metal-oxide semiconductor field-effect transistor (MOSFET) designated as S, a diode labeled Do, an inductor represented by L, a filter capacitor denoted as C, and a load resistor RL. The switch operates ON and OFF at a frequency fs and has a duty cycle of D.

1.2 DC Model

$M_{VDC}=\frac{Vo}{VI}=D,$

and the DC current transfer function of the ideal converter is

$M_{IDC}=\frac{Io}{I_I}=\frac{1}{D}$

As shown in Figure 2, the waveform of the switch current can be approximated by $i_s = \begin{matrix} I_L DT<t\leq DT \\ 0 \quad DT<t\leq T \end{matrix}$

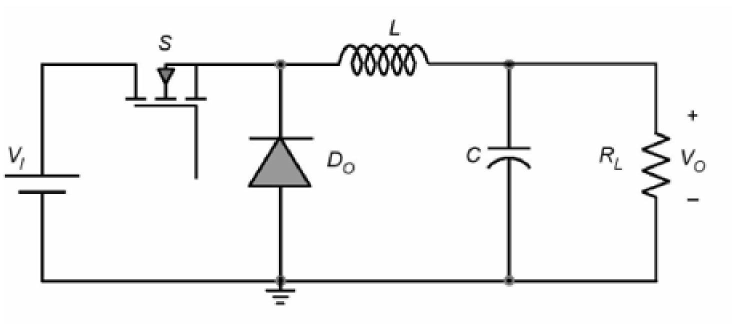


Fig 1 Circuit of the buck converter.

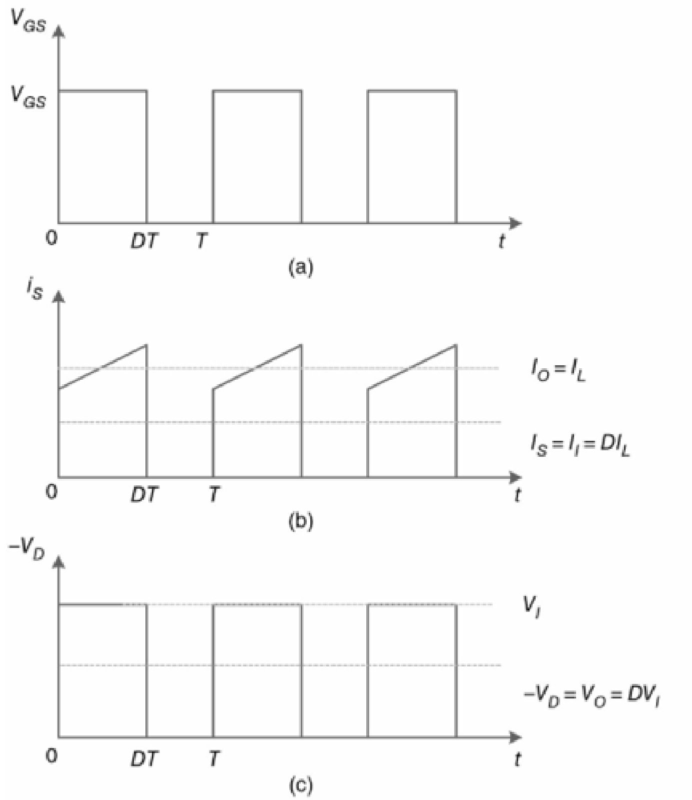


Fig 2 Waveforms of the ideal switching network. (a) Gate-to-source voltage of switch. (b) Switch current. (c) Diode voltage. *DT*

The DC component of the switch current is

$$I_s = \frac{1}{T}\int_0^T i_s dt = \frac{1}{T}\int_0^{DT} i_L dt = DI_L$$

Hence, the MOSFET can be replaced by an ideal DC current-dependent current source.

As illustrated in Figure 2.2c, the waveform of the diode inverted voltage, equal to the voltage between the inductor and diode terminals, is given by

$$-v_D(t) = v_{KA}(t) = v_{LD} = \begin{cases} V_I \; for \; 0 < t \leq DT \\ 0 \; for \; DT < t \leq T \end{cases}$$

The DC component of diode inverted voltage is

$$-V_D = V_{KA} = V_{LD} = \frac{1}{T}\int_0^T V_{KA} dt = \frac{1}{T}\int_0^{DT} V_{SD} dt = DV_{SD} = DV_I,$$

where V_{KA} is the DC component of diode branch voltage between the cathode and anode terminals, V_{SD} is the DC component of voltage between the drain of the MOSFET and anode of the diode, and V_{LD} is the DC component of voltage between the inductor (positive terminal) and the anode of the diode. The diode can be

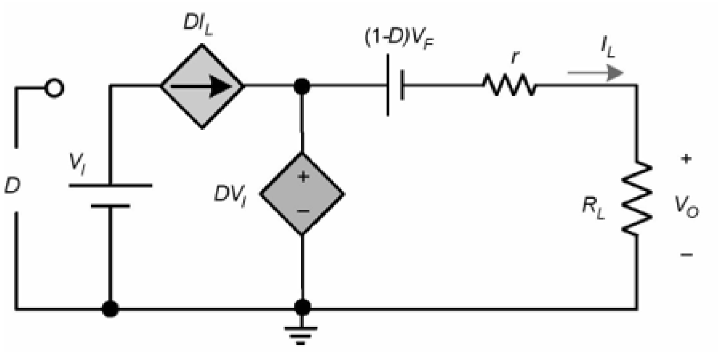


Fig 3 DC model of buck DC–DC converter in CCM.

substituted with an ideal DC voltage-dependent voltage source. The averaged DC model of the buck converter operating in Continuous Conduction Mode (CCM) is illustrated in Figure 3. In this DC model, the inductor is represented as a short circuit, while the filter capacitor is treated as an open circuit. This model captures the converter's performance during steady-state conditions. The DC component of the input current is

$$I_I = I_S = DI_L$$

$$I_O = I_L.$$

Using the conservation of energy principle [37, 45], the series resistance r that corresponds to the inductance L is equal to the total averaged resistance of the switch, diode, and inductor section.

$$r = Dr_{DS} + (1 - D)R_F + r_L,$$

where r_L, r_{DS}, and R_F are the equivalent series resistance (ESR) of the inductor, ON-state resistance of the MOSFET, and diode forward resistance, respectively. The resistance r produces the same loss in the form of heat as that produced by the individual resistances placed in their respective branches. The efficiency of the converter is

$$\eta = \frac{Po}{P_I} = \frac{V_o I_o}{V_I I_I} = M_{IDC} M_{VDC}$$

$$= \frac{1}{1 + \frac{Dr_{DS} + (1 - D)R_F + r_L}{R_L} + \frac{(1 - D)V_F}{V_o} + \frac{f_s C_o R_L}{{M_{VDC}}^2} + \frac{r_L R_F (1 - D)^2}{12 {f_s}^2 L^2}}$$

where r_L,r_{DS}, V_F, R_F, r_C, and C_o are the ESR of the inductor, ON-state resistance of the MOSFET, diode forward voltage drop, diode forward resistance, ESR of filter capacitor, and MOSFET output capacitance, respectively.

1.3 Design Example

The subsequent analysis is done for the buck converter designed with the following specifications:

i Supply voltage $V_I = 10$ V.

ii Output voltage $V_O = 6$ V.

iii Load resistance $R_L = 5\ \Omega$.

iv Switching frequency $f_s = 150$ kHz.

v The inductance to ensure CCM operation is $L = 20\ \mu$H.

vi The capacitance to ensure CCM operation is $C = 330\ \mu$F.

The circuit parameters include an inductor ESR (rL) of 0.2 Ω and a capacitor ESR (rC of 0.1 Ω. Regarding the semiconductor components, the MOSFET's ON-state resistance is rDS=0.1 Ω, and the diode's forward resistance is RF=0.03 Ω. To maintain the rated output voltage, the system operates at a duty cycle (D) of 0.54. Utilizing equation (2.9), the resulting equivalent averaged resistance (r) is approximately 0.27 Ω.

2 Large-Signal and Small-Signal Models of Buck Converter

Apply perturbation and averaging technique :

$$i_S = I_S + i_s,$$

$$i_L = I_L + i_l,$$

$$v_D = V_D + v_d,$$

$$v_I = V_I + v_i,$$

$$v_O = V_O + v_o,$$

$$d_T = D + d, and$$

where i_S, i_L, v_D, v_I, v_O, and d_T are the large-signal parameter . and this i_s, i_l, v_d, v_i, v_o, and d are the small-signal parameter of buck converter , respectively. Hence

$$i_S = d_T i_L = (D + d)(I_L + i_l) = DI_L + Di_l + I_L d + i_l d$$

and

$$-v_D = d_T v_I = (D + d)(V_I + v_i) = DV_I + Dv_i + V_I d + v_i d.$$

The large-signal nonlinear model of a buck converter in CCM is shown in Figure 2.4.

$$i_S = DI_L + Di_l + I_L d$$

and

$$-v_D = DV_I + Dv_i + V_I d.$$

the large-signal linear model of the buck converter.The AC terms are separated from DC components to obtain the linear, averaged, small-signal model as shown in Figure 4.

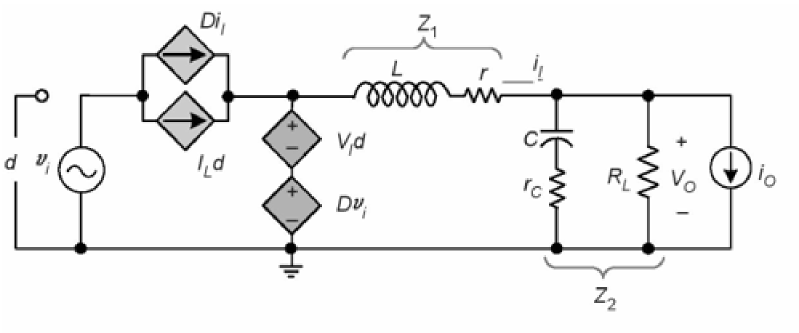


Fig 4 Small-signal model of the PWM buck DC–DC converter in CCM.

3 Power Stage Transfer Functions

The small-signal model of the buck converter

The impedances Z_1 and Z_2 are

$Z_1 = r + sL$

And

$$Z_2 = R_L || \left(r_c + \frac{1}{sC}\right) = \frac{R_L\left(r_C + \frac{1}{sC}\right)}{\left(R_L + r_C + \frac{1}{sC}\right)}$$

The current through the parallel combination of the load resistance and the filter capacitor is

$$i_{Z_2} = \frac{v_o}{Z_2}$$

The current through the inductor is

$$i_l = i_{Z_2} + i_o = \frac{v_o}{Z_2} + i_o$$

The input and the switch currents are

$$i_i = i_s = Di_l + I_L d.$$

Applying Kirchhoff's voltage law, we obtain

$v_o = Dv_i + V_I d - i_l Z_1.$

Substituting (2.26) in (2.24), we get

$$i_l = \frac{Dv_i + V_I d - Z_1 i_l}{Z_2} + i_o = \frac{Dv_i + V_I d + i_0}{Z_1 + Z_2}$$

Applying Kirchhoff's voltage law, the output voltage is

$$i_l = (Dv_i + V_I d)\frac{Z_2}{Z_2 + Z_1}$$

3.1 Duty Cycle-to-Output Voltage Transfer Function T_p

The duty cycle-to-output voltage transfer function is obtained by setting $v_i = 0$ and $i_o = 0$ in Figure 2.6. Setting $v_i = 0$ in (2.28) and rearranging, we get

$$v_o = (V_I d)\frac{Z_2}{Z_2 + Z_1}$$

Hence, the duty cycle-to-output transfer function is

$$T_p(s) = \frac{v_{0(s)}}{d(s)}|_{v_i = i_0 = 0} = (V_I)\frac{Z_2}{Z_2 + Z_1}$$

Substituting(2.21)and(2.22)into(2.30)givesthecontrol-to-outputtransferfunction in the s domain as

$$T_p(s) = \frac{v_{o(s)}}{d(s)}|_{v_i = i_0 = 0} = T_{px}\frac{s + \omega_{zn}}{s^2 + 2\xi\omega_0 s + \omega_0^2} = T_{po}\frac{1 + \frac{s}{\omega_{zn}}}{\left(\frac{S}{w_0}\right)^2 + \frac{2\xi S}{w_0} + 1}$$

Where

$$T_{px} = \frac{V_I R_L r_c}{L(R_L + r_c)}$$

the DC gain T_{po} is

$$T_{px} = T_p(0) = \frac{V_I R_L}{(R_L + r)} \;,$$

the angular corner frequency or the angular undamped natural frequency is

$$\omega_0 = \sqrt{\frac{R_L + r}{LC(R + r_c)}}$$

the damping ratio is

$$\xi = \frac{L + clR_l(r_c + r) + \gamma_{c^*} r}{2\sqrt{LC(R_L + r_c)(R_L + r)^1}}$$

and the angular frequency of the left-half plane (LHP) zero is

$$\omega_{zn} = \frac{1}{r_c C}$$

The magnitude of the transfer function is

$$\left|T_p\right| = T_{po}\sqrt{\frac{\left[1+\left(\frac{w}{w_{zn}}\right)^2\right]}{\left[1-\left(\frac{w}{w_{zn}}\right)^2\right]^2+4\xi^2\left(\frac{w}{\omega_0}\right)^2}}$$

and the phase of the transfer function is

$$\phi_{T_p} = -180^0 + \tan^{-1}\left(\frac{w}{w_{zn}}\right) - \tan^{-1}\left[\frac{2\xi}{1-\left(\frac{w}{w_{zn}}\right)^2}\right]$$

For converters, whose damping coefficient is closer to unity, the converter bandwidth f_{BW} is nearly equal to the natural frequency f_o. In contrast, for transfer functions with a lower ξ, the bandwidth is higher than the natural frequency. The 3-dB bandwidth of the converter power stage determines the speed of the outer-voltage loop in voltage-mode control. The gain at f_{BW} is

$$\frac{|T_p(f_{B\omega})|}{T_{po}} = \frac{1}{\sqrt{2}}$$

The following analysis for the transfer function bandwidth holds true for $\omega_{zn} >> \omega_o$. The bandwidth of the T_p transfer function is governed by the dominant poles and assuming dominant poles,

$$\frac{|T_p(f_{B\omega})|}{T_{po}} = \frac{1}{\sqrt{2}} = \sqrt{\frac{1}{\left[1-\left(\frac{w_{B\omega}}{w_o}\right)^2\right]^2+4\xi^2\left(\frac{w_{B\omega}}{\omega_0}\right)^2}}\ ,$$

Yielding

$$\frac{1}{\left[1-\left(\frac{f_{B\omega}}{f_o}\right)^2\right]^2 + 4\xi^2\left(\frac{f_{B\omega}}{f_o}\right)^2} = \frac{1}{2}$$

and producing the bandwidth as

$$f_{B\omega} = f_0\sqrt{1 - 2\xi^2 + \sqrt{4\xi^4 - 2\xi^2 + 2}}$$

For the buck converter designed in Section 1.3, the DC value of duty cycle-tooutput voltage transfer function is $T_{po} = 8.81$ V = 18.89 dB V, natural corner frequency is $f_o = \omega_o/2\pi$ = 2.07 kHz, frequency of the LHP zero is $f_{zn} = \omega_{zn}/2\pi = 4.82$ kHz, and the damping coefficient is $\xi = 0.4$. The gain decreases by 40 dB/dec beyond f_o. The presence of a zero frequency is attributed to the series connection of the filter capacitor and its internal equivalent series resistance (ESR). At DC and low-frequency ranges, the phase remains at 0∘, signifying a direct relationship where the output voltage follows the fluctuations of the duty cycle. However, at higher frequencies, the zero frequency (f_{zn}) causes the phase to level out at −90∘. A phase dip is observed reaching a minimum of −160∘ at 3.2 kHz. This phase behavior is highly sensitive to the load resistance (RL); a reduction in R_L pushes the phase closer to −180∘ and decreases the damping coefficient, which subsequently triggers ringing in the output voltage at frequency fo. These characteristics are visually represented in the magnitude and phase plots of the transfer function in Figure 2.7.

3.2 Duty Cycle-to-Inductor Current Transfer Function T_{pi}

The duty cycle-to-inductor current transfer function is obtained by setting $v_i = 0$ and $i_o = 0$ in Figure 2.6. Setting $v_i = 0$ in (2.24) and rearranging, we get

$$i_l = \frac{V_I d}{Z_2 + Z_1}$$

Hence, the duty cycle-to-inductor current transfer function is

$$T_{pi}(s) = \frac{i_{l(s)}}{d(s)}\Big|_{v_i=i_0=0} = \frac{V_I}{Z_2 + Z_1}$$

Substituting (2.21) and (2.22) into (2.44) gives the duty cycle-to-inductor current transfer function in the s-domain as

$$T_{pi}(s) = \left.\frac{i_l(s)}{d(s)}\right|_{v_i=i_o=0} = T_{pix}\frac{s + \omega_{zi}}{s^2 + 2\xi\omega_o s + \omega_o^2} = T_{pio}\frac{1 + \dfrac{s}{\omega_{zi}}}{\left(\dfrac{s}{\omega_o}\right)^2 + \dfrac{2\xi s}{\omega_o} + 1},$$

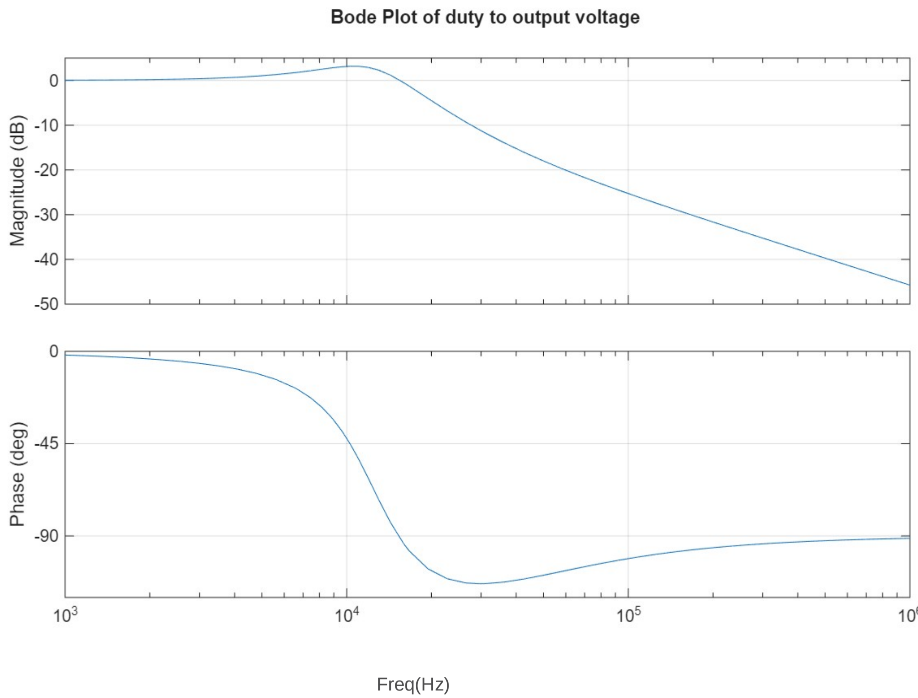


Fig 5 plots of duty cycle-to-output voltage transfer function T_p

where

$$T_{pix} = \frac{V_I}{L}$$

the DC gain T_{pio} is

$$T_{pio} = T_{pi}(0) = \frac{V_I}{(R_L + r)} \quad ,$$

and the angular freq of the LHP zero is

$$w_{zi} = \frac{1}{c(R_L + r_c)}$$

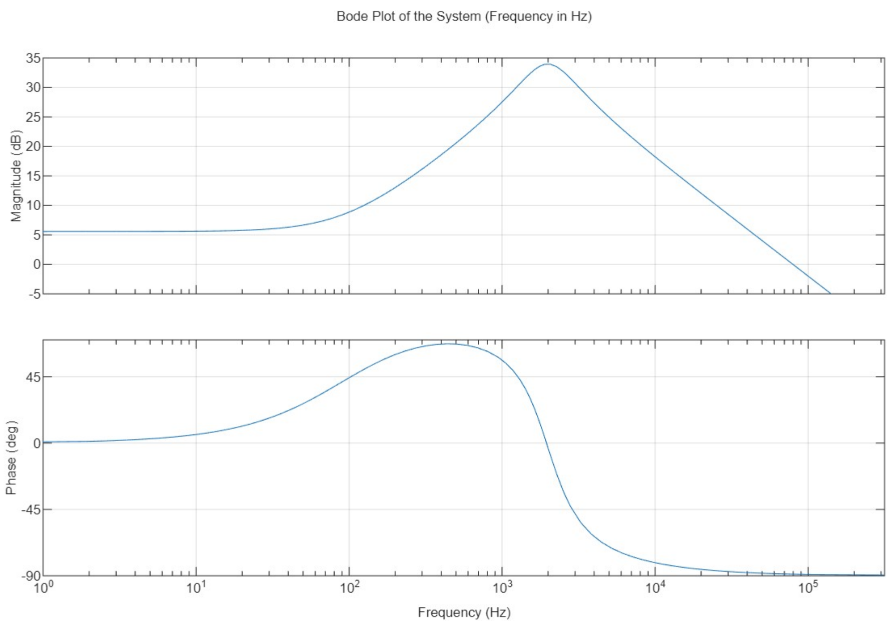
The magnitude of the duty cycle-to-inductor current transfer function is

$$|T_{pi}| = T_{pio}\sqrt{\frac{\left[1+\left(\frac{w}{w_{zi}}\right)^2\right]}{\left[1-\left(\frac{w}{w_o}\right)^2\right]^2+4\xi^2\left(\frac{w}{\omega_0}\right)^2}}$$

and the phase of the transfer function is

$$\phi_{T_{pi}} = -180^0 + \tan^{-1}(\frac{w}{w_{zi}}) - \tan^{-1}[\frac{2\xi\frac{w}{w_{ZO}}}{1-(\frac{w}{w_{ZO}})^2}]$$

Figure 6 shows the theoretically obtained magnitude and phase plots of duty cycle-to-inductor current transfer function. The gain at DC and low frequencies is



(b)

Fig 6 plots of duty cycle-to-inductor current transfer function T_{pi}

The DC gain of the duty cycle-to-inductor current transfer function is approximately T_{pio}= 1.9 A (5.58 dBA). The LC filter formed by the inductor, capacitor, and load resistance behaves as a second-order system with a natural frequency of f_o = 2.07 kHz. In addition, a low-frequency left-half-plane (LHP) zero occurs at f_{zi}= 94.68Hz.

Beyond the corner frequency (f_o), the magnitude of the transfer function decreases at a slope of 20 dB per decade. The presence of the low-frequency zero f_{zi} and the complex-conjugate poles at (f_o) characterizes the dynamic behavior of the system. The phase response begins at 0° and increases to approximately 45° due to the effect of the zero, indicating that at low frequencies the duty cycle variation leads the inductor current.

Compared to the control-to-output transfer function (T_p), the bandwidth of (Tpi) is larger, resulting in a faster response of the inductor current to variations in duty cycle. The location of the zero (f_{zi}) is influenced by the load resistance (R_L) and the capacitor parameters (C) and (r_C). As the load resistance decreases while keeping the capacitance constant, the zero frequency shifts closer to the natural frequency (f_o).

At a particular minimum value of load resistance, the zero frequency (f_{zi}) approaches (f_o), effectively canceling one of the complex-conjugate poles. This leads to a transition from a second-order system to a first-order dominant response. In this condition, the dominant pole is primarily determined by the inductor and the load resistance.

Table 2.1 Summary of O-L transfer functions.

Variable	Value
$\lvert T_{po} \rvert$	8.81 V = 18.89 dBV
$\lvert T_{pio} \rvert$	1.9 A = 5.58 dBA
$\lvert M_{vo} \rvert$	0.6 V/V =−4.4 dB
$\lvert M_{vio} \rvert$	0.054 A/V =−25.35 dBA/V
ω_o	13 krad/s
f_o	2.07 kHz
ξ	0.4
ω_{zi}	0.594 krad/s
f_{zi}	94.57 Hz
ω_{zn}	30.284 krad/s
f_{zn}	4.82 kHz

Under steady-state (DC) conditions, the inductor behaves as a short circuit, while the output capacitor effectively acts as an open circuit. Consequently, the equivalent resistance seen at the output is determined by the parallel combination of the load resistance (RL) and the averaged resistance (r).

As the operating frequency increases, the capacitive reactance initially remains large, and the system behavior is primarily influenced by the inductive reactance. At the resonant frequency (f_o), the magnitudes of the inductive and capacitive reactances become equal, causing resonance in the filter network of the converter.

For frequencies higher than (f_o), the capacitive reactance becomes dominant, leading to a significant reduction in the output impedance, approaching a very small value. The phase response begins at 0° and rises toward 90° due to the presence of a zero at (w_{ri}). Around the pole frequencies, the phase shifts downward toward −90°. However, the left-half-plane zero at (w_{zn}) contributes to an increase in phase, bringing it back toward 0° at higher frequencies.

A summary of the important open-loop characteristics and parameters is presented in Table 2.1.

4 Inner-Current Loop

The structure of the inner current control loop, including the low-pass filter in the feedback path, is illustrated in Figure7, while the corresponding circuit implementation is presented in Figure 2.16. The inductor current is measured using a sensing resistor ($R_s=1\Omega$), which is assumed to have a value of for the analysis.A low-pass filter is incorporated in the current sensing path to suppress high-frequency components arising from switching actions, thereby ensuring that only the average value of the inductor current is utilized for control purposes.

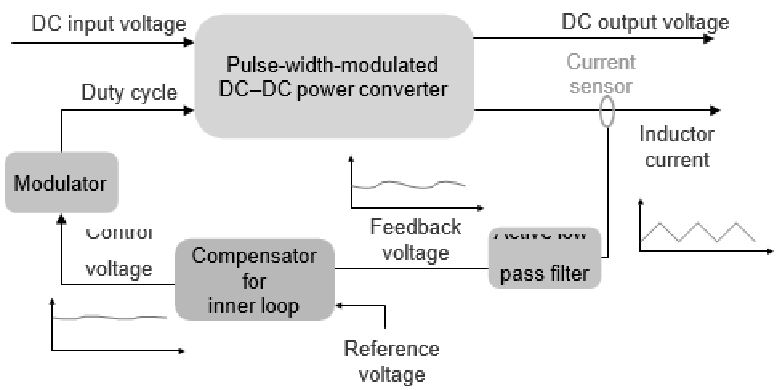


Fig 7 Block diagram of the inner-current loop with low-pass filter in the feedback .

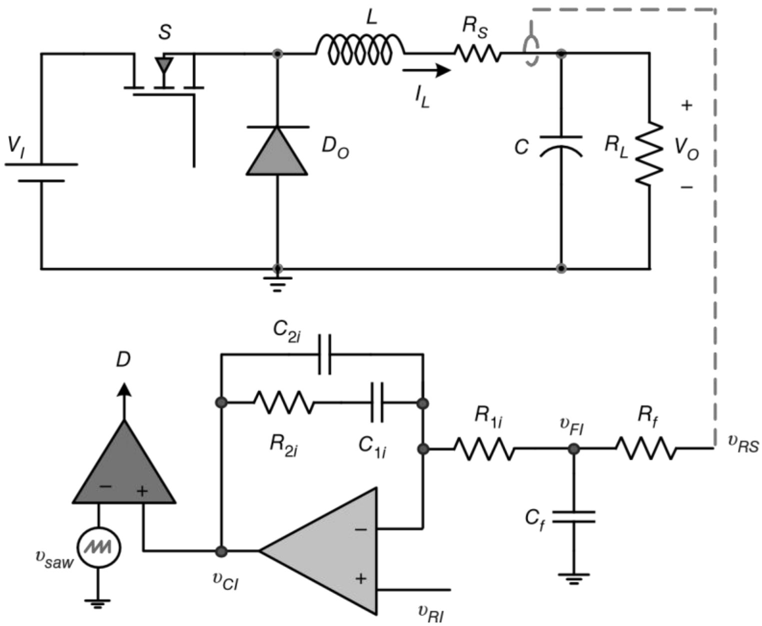


Fig 8 Circuit of the buck DC–DC converter with inner-current loop.

its harmonics, which are present in the sensed voltage $v_{RS} = R_s i_L$. Therefore, the output voltage obtain of the low-pass filter is $V_{FI}=R_s I_L$ and is governedbytheDCcomponentofthesensedinductorcurrent.Thus,in this method,the *average* component of the inductor current is sensed, thereby reducing the problems due to the subharmonic instability and transistor misgating. The transfer functions related to the inner-current loop, which include filter circuit in the feedback path, are derived and discussed in this section.

4.1 Transfer Function of Filter and Non-inverting Amplifier T_f

The active low-pass filter contain resistor R_f and a capacitor C_f. The transfer function of filter is

$$T_f(s) = \frac{v_{fi}(s)}{v_{rs}(s)} = \frac{Z_{cf}}{R_f + Z_{cf}} = \frac{1/sC_f}{R_f + \frac{1}{sC_f}} = \frac{1}{sR_f C_f + 1}$$

$$\frac{1}{R_f C_f} \frac{1}{(s + \frac{1}{R_f C_f})} = \frac{w_{pf}}{s + w_{pf}} = \frac{1}{1 + \frac{s}{w_{pf}}}$$

Setting $s = j\omega$,

$$T_f(jw) = \frac{1}{1 + \frac{jw}{w_{pf}}}$$

yielding

$$T_f(f) = \frac{1}{1+\frac{jf}{f_{pf}}} = |T_f(f)|e^{j\phi(f)}$$

where

$$|T_f(f)| = \frac{1}{\sqrt{1+\left(\frac{f}{f_{pf}}\right)^2}}$$

and

$$\phi(F) = tan^{-1}\left(\frac{f}{f_{pf}}\right)$$

$$\left(\frac{f_{pf}}{f}\right) = \sqrt{\frac{|T_f(f)|^2}{1-|T_f(f)|^2}}$$

Hence,

$$f_{pf} = f\sqrt{\frac{|T_f(f)|^2}{1-|T_f(f)|^2}}$$

The magnitude of the transfer function T_f in terms of the filter input and output voltages is also given as

$$|T_f(f)| = \frac{\Delta V_{fI}}{\Delta V_{RS}} \quad .$$

The filter frequency can be set by amount of attenuation desired. As an example, to achieve an attenuation of 50% for $f = f_s = 150$ kHz, the filter upper cutoff frequency can be calculated using (2. 89) as

$$f_{pf} = f_s\sqrt{\frac{|T_f(f)|^2}{1-|T_f(f)|^2}}$$

4.2 Transfer Function of Pulse-Width Modulator T_m

control voltage-to-duty cycle transfer function of the PWM is

$$T_m = \frac{D}{V_c} = \frac{d}{V_c} = \frac{1}{V_m}$$

where V_{Tm} is the peak voltage of sawtooth waveform.

The steady-state average inductor current $I_L = V_O/R_L = 1.2$A. At a duty cycle $D = 0.54$ and assuming the peak sawtooth voltage of the PWM is

V_{Tm} = 1.88 V, the required steady-state control voltage is $V_{CI} = DV_{Tm} = 0.54 \times 1.88 =$ 1.02V.

4.3 Uncompensated Loop Gain T_{ki}

The behavior of the inner-current loop can be analys by using the uncompensated loop gain is .

$$T_{ki} = \frac{v_{fi}}{v_{ei}} = T_m T_{pi} R_s T_f = T_{kio} \frac{1 + \frac{s}{\omega_{zi}}}{(1 + \frac{s}{w_{pf}})[\left(\frac{S}{w_0}\right)^2 + \frac{2\xi S}{w_0} + 1]}$$

$$T_{ki} = \frac{v_{fi}}{v_{ei}} = T_m T_{pi} R_s T_f = T_{kio} \frac{1+\frac{s}{\omega_{zi}}}{(1+\frac{s}{w_{pf}})[\left(\frac{S}{w_0}\right)^2 + \frac{2\xi S}{w_0} + 1]}$$

where T_m, T_{pi}, and T_f . The value of sense resistor is $R_s = 0.5\ \Omega$. The DC gain T_{kio} is given by

$$T_{ki0} = \frac{R_S V_i}{V_m R_L + r}$$

The gain at DC is $T_{kio} = 1.08 = 0.67$ dB. A sufficiently large DC gain is essential to minimize steady-state deviations in the output voltage caused by variations in input voltage or load conditions. Therefore, the control system should be designed to ensure high gain at low frequencies while also providing adequate phase enhancement near the crossover frequency of the loop gain. To meet these requirements, a single-lead integral (Type-II) compensator is implemented for the inner current loop, as discussed in Section 4.4.

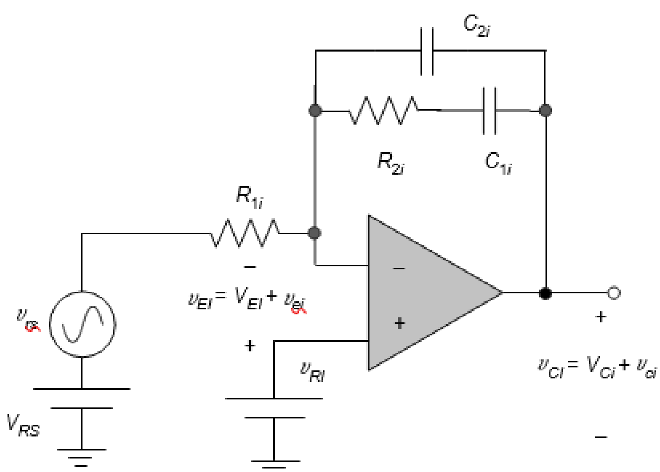


Fig 9 Circuit of Type-II controller.

4.4 Transfer Function of Control Circuit for Inner-Current Loop T_{ci}

Figure 9 illustrates the configuration of a single-lead integral controller, commonly referred to as a Type-II compensator. This controller introduces a pole at the origin to increase the low-frequency (DC) gain, along with a pole–zero pair that enables adjustment of the crossover frequency (f_c) in order to achieve the desired phase margin (PM).

A systematic method for selecting the controller parameters to obtain sufficient DC gain (greater than 20 dB) and to ensure a phase margin of approximately (60 degree) for the inner current loop is discussed in the literature [45, 51]. The subsequent section focuses on the detailed design of this Type-II controller for the inner-current loop.

The magnitude of the uncompensated loop gain of transfer function can be find using

$$|T_{ki}(f)| = T_{ki0}\frac{\sqrt{1+\left(\frac{f}{f_{zi}}\right)^2}}{\sqrt{1+\left(\frac{f}{f_{pi}}\right)^2}\sqrt{\left(1-\left(\frac{f}{f_0}\right)^2\right)^2+\left(2\zeta\frac{f}{f_0}\right)^2}}$$

and the phase is

$$\phi_{Tki}(f) = -180^\circ + \tan^{-1}\left(\frac{f}{f_{zi}}\right) - \tan^{-1}\left(\frac{f}{f_{pi}}\right) - \tan^{-1}\left(\frac{2\zeta\frac{f}{f_0}}{1-\left(\frac{f}{f_0}\right)^2}\right)$$

The desired crossover frequency(f_c) for the loop gain is chosen as f_c = 10 kHz. At this frequency, the magnitude of the uncompensated loop gain

$|T_{ki}(f_c)|$ = 0.583 = −4.68 dB, and its phase is $\phi_{Tk}(f_c) = -98^\circ$. The required phase margin is $PM = 60^\circ$. Therefore, the phase boost required at f_c is

$$\phi_m = PM - \phi_{Tk}(f_c) - 90^\circ = 68^\circ,$$

yielding the geometric mean between the controller corner frequencies as [47]

$$K = \tan\left(\frac{\phi_m}{2} + 45^\circ\right) = 5.8$$

Let R_{1i} = 1 kΩ⁄0.25 W⁄2%, the value of C_{2i} is

$$C_{2i} = \frac{|T_{ki}(f_c)|}{2\pi f_c K R_{1i}} =$$

A 1.6 nF⁄25 V standard capacitor is selected. The expression for C_{1i} is

$$C_{1i} = C_{2i}(K^2 - 1) = 51 \text{ nF}.$$

A 51 nF⁄25 V standard capacitor is selected. The expression for R_{2i} is

$$R_{2i} = \frac{K}{2\pi f_c C_{1i}} = 1.8 \text{ k}\Omega.$$

A 1.8 kΩ⁄0.25 W⁄2% standard resistor is selected. The transfer function T_{ci} of the current-loop control circuit is

$$T_c(s) = \frac{v_c(s)}{v_{ei}(s)} = T_{ci0}\frac{1+\frac{s}{\omega_{zc}}}{s\left(1+\frac{s}{\omega_{pc}}\right)}$$

where

,

$$T_{ci0} = \frac{1}{R_{1i}(C_{1i}+C_{2i})} = 19kV/V,$$

$$\omega_{zc} = \frac{1}{R_{2i}C_{1i}} = 11krad/s$$

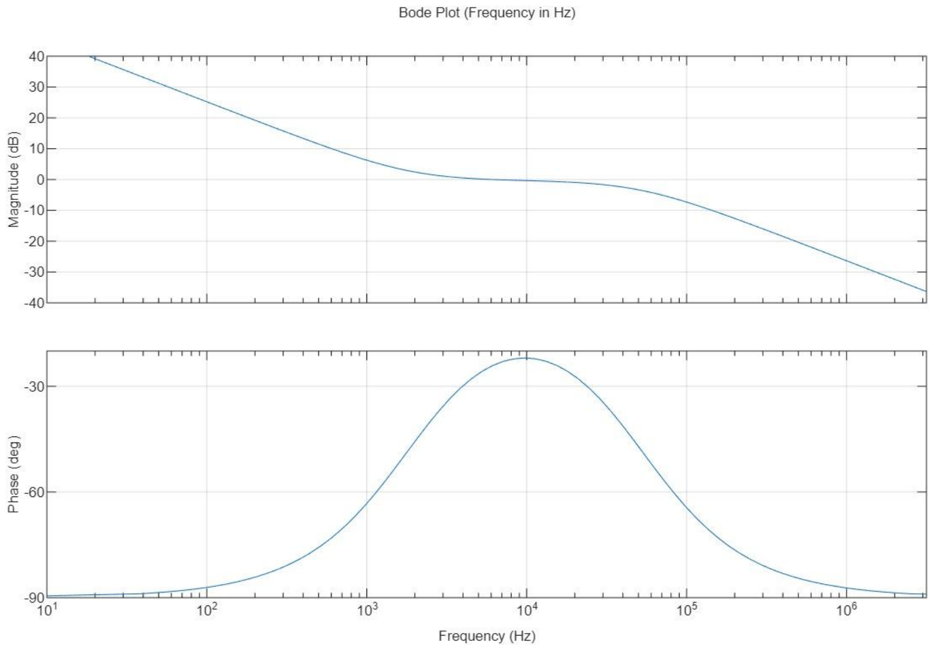


Fig 10 plots of controller transfer function T_{ci}.

and

$$\omega_{pc} = \frac{C_{1i}+C_{2i}}{R_{2i}C_{1i}C_{2i}} = 358 \text{ krad/s}.$$

Figure 10 presents the theoretical magnitude and phase responses of the designed Type-II controller utilized in the inner current loop. The corresponding control transfer function (T_{ci}) is included in the main signal path to evaluate and establish the overall inner-loop gain.

4.5 Compensated Loop Gain of Inner-Current Loop T_i

The loop gain of the compensated inner-current loop is

$$T_i = \frac{v_i}{v_{ei}} = T_{ki}T_c = T_m T_{pi} R_s T_f T_c$$

Finally the transfer function of inner current loop is .

$$T_i = T_{io}\frac{(1+\frac{s}{\omega_{zi}})(1+\frac{s}{\omega_{zci}})}{s(1+\frac{s}{w_{pf}})(1+\frac{s}{\omega_{pci}})[\left(\frac{S}{w_0}\right)^2+\frac{2\xi S}{w_0}+1]}$$

where T_{io} is the gain at $s = 0$ given as

$$T_{i0} = T_{ki0}T_{ci0} = \frac{V_i R_s}{V_m(R_L + r)} \cdot \frac{1}{R_{1i}(C_{1i} + C_{2i})}$$

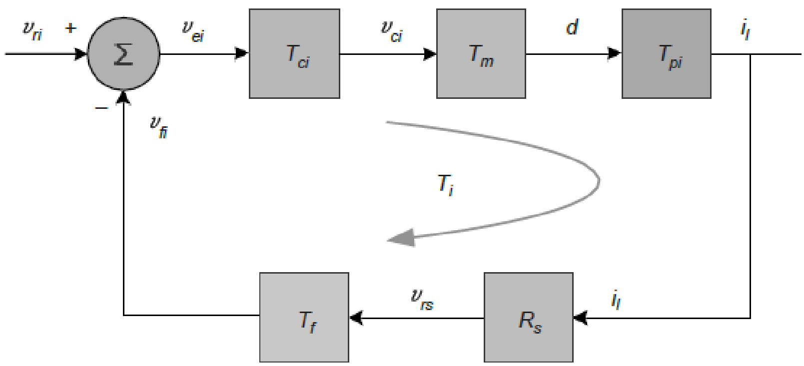
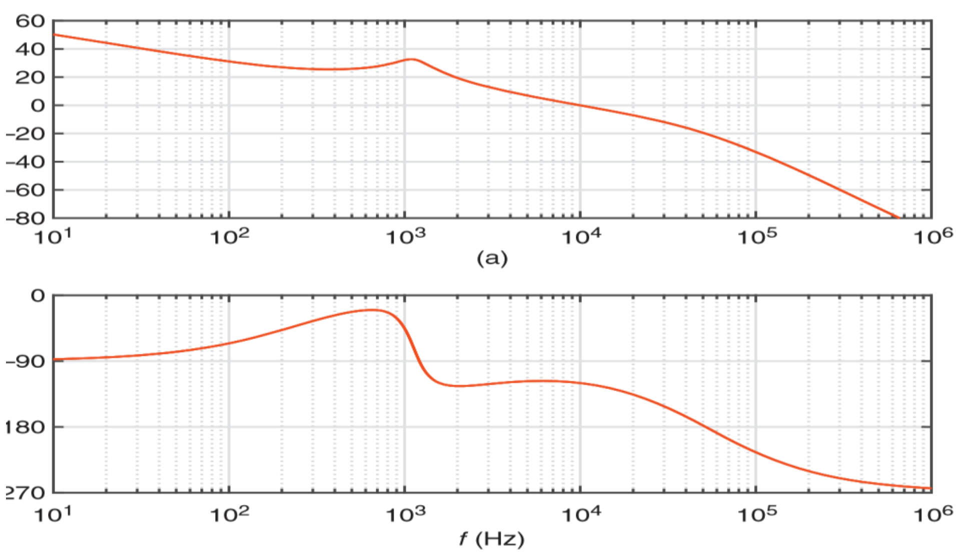


Fig 11



(b)

Fig 12 plots of compensated loop gain T_i of the inner current loop.

The calculated value of compensated loop gain at DC is $T_{io} > 50$ dB. The phase margin is $PM = 60^{\circ}$ and the gain marginis 20 dB. The closed-loop transfer functions for the inner-current loop are derived .

.

5 Closed-Loop Transfer Functions for the Inner-Current Loop

The small-signal current reference v_{ri} for the inner loop is given by the outer loop. This section discusses only the inner-current loop without the presence of outer-voltage loop. The inner-current loop in the presence of outer-voltage loop .

5.1 Reference Voltage-to-Inductor Current Transfer Function T_{icl}

The closed-loop reference voltage-to-inductor current transfer function is

$$T_{i,cl}(s) = \frac{i_l(s)}{v_{ri}(s)}|_{v_i=i_o=0} = \frac{T_{ci}T_mT_{pi}}{1+T_i}$$

5.2 Reference Voltage-to-Output Voltage Transfer Function T_{picl}

The closed-loop reference voltageto-output voltage transfer function is

$$T_{v,cl}(s) = \frac{v_o(s)}{v_{ri}(s)} = \frac{T_{ci}T_mT_{pi}T_z}{1+T_i}$$

where T_z is the inductor current-to-output voltage transfer function given is :

$$T_z(s) = \frac{v_o(s)}{i_l(s)} = Z_o = R_L \parallel \left(r_c + \frac{1}{sC}\right) = \frac{R_L\left(r_c+\frac{1}{sC}\right)}{R_L+r_c+\frac{1}{sC}}$$

5.3 Input Voltage-to-Inductor Current Transfer Function M_{icl}

The closed-loop input voltage to inductor current transfer function is

$$M_{icl}(s) = \frac{i_l(s)}{v_i(s)}|_{v_{ri}=i_o=0}$$

From the diagram,

$$i_l = i_l' + i_l'' = -T_m T_{pi} R_s T_f i_l + M_{vi} v_i = -T_i i_l + M_{vi} v_i$$

$$i_l(1 + T_i) = M_{vi} v_i$$

$$M_{icl}(s) = \frac{i_l(s)}{v_i(s)} = \frac{M_{vi}}{1+T_i}$$

5.4 Input Voltage-to-Output Voltage Transfer Function M_{vicl}

The closed-loop input voltage-to output voltage transfer function is

$$M_{vid}(s) = \frac{v_o(s)}{v_i(s)}\Big|_{v_{ri}=i_o=0}$$

The closed-loop input-to-output transfer function is

$$M_{vicl}(s) = T_z M_{icl} = T_z \frac{M_{vi}}{1+T_i} \quad .$$

6 Outer-Voltage Loop

To ensure a regulated and stable output voltage, an outer voltage control loop is incorporated, which operates using a reference voltage (v_{RV}}). In a dual-loop control structure, this reference signal is typically derived as a scaled portion of the desired output voltage. The scaling factor is defined by the feedback network, whose gain (β) is set using a resistor divider arrangement.The overall configuration of the average current-mode controlled buck converter with the outer voltage loop is depicted in Figure 13, while the corresponding circuit implementation is shown in Figure 14.

6.1 Transfer Function of Feedback Network β

The transfer function of the feedback network is given as

$$\beta = \frac{v_{fb}}{v_o} \quad ,$$

where V_{RV} is the reference voltage to the outer-voltage loop. Assuming the reference voltage V_{RV} = 1.88 V, the transfer function of the feedback network is

$$\beta = \frac{V_{RV}}{V_o} = \frac{6}{6} = 1$$

6.2 Uncompensated Loop Gain for Outer-Voltage Loop T_{kv}

The natural behavior of the outer-voltage loop can be determined using the uncompensated loop gain as

$$T_{kv} = \frac{v_{fv}}{v_{ev}} = \frac{\beta v_o}{v_{ri}} = \beta T_{picl}.$$

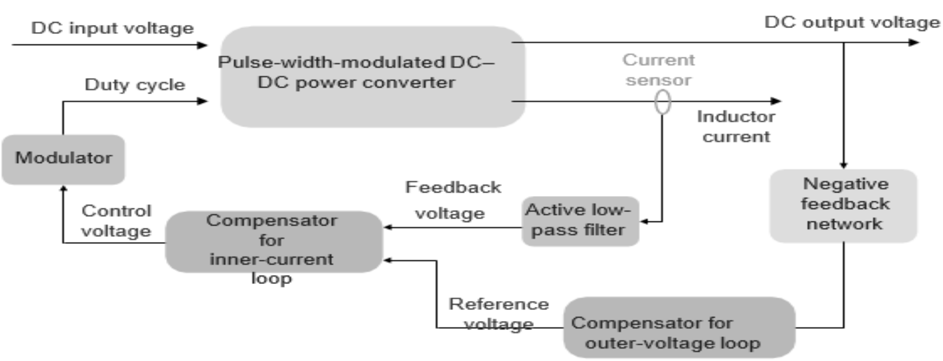


Figure 13 Block diagram of average current-mode controlled buck converter with outer-voltage loop.

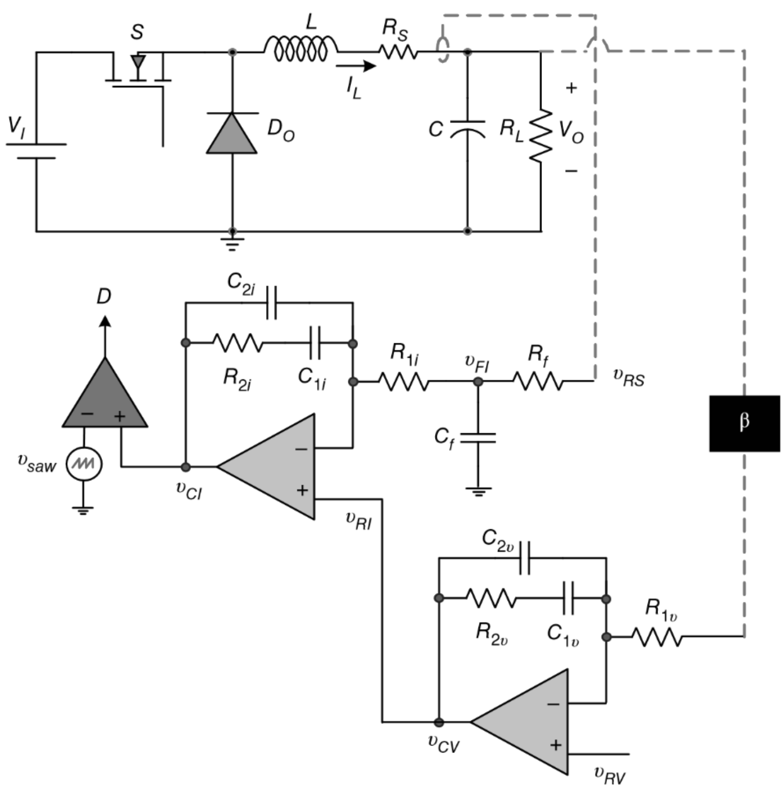


Fig 14 Circuit of the average current-mode controlled buck converter with outer-voltage loop.

6.3 Transfer Function of Control Circuit for Outer-Voltage Loop T_{cv}

Figure 2.36 depicts the implementation of a Type-II compensator applied in the outer voltage control loop. This controller provides a pole at the origin to enhance the low-frequency (DC) gain, along with a pole–zero combination that enables tuning of the crossover frequency (f_c) to achieve the desired phase margin (PM).A systematic design approach is used to select the controller parameters such that the DC gain exceeds 20 dB while ensuring an adequate phase margin of approximately (60 degree) for stable loop operation

The desired crossover frequency for the loop gain is chosen as f_c = 2 kHz. At this frequency, the magnitude of the uncompensated loop gain transfer function is $|T_k(f_c)| = 0.22$ V/V = −13.1 dB and its phase is $\phi_{Tkv}(f_c) = -81^\circ$. The required phase margin is $PM = 60^\circ$. Therefore, the phase boost required at f_c is $\phi_m = PM - \phi_{Tkv}(f_c) - 90^\circ = 51^\circ$,

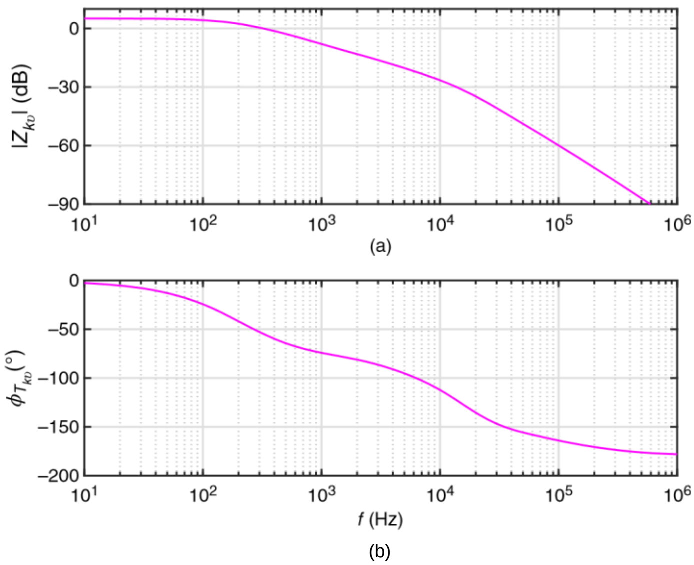


Fig 15 Theoretically obtained plots of uncompensated loop gain T_{kv}.

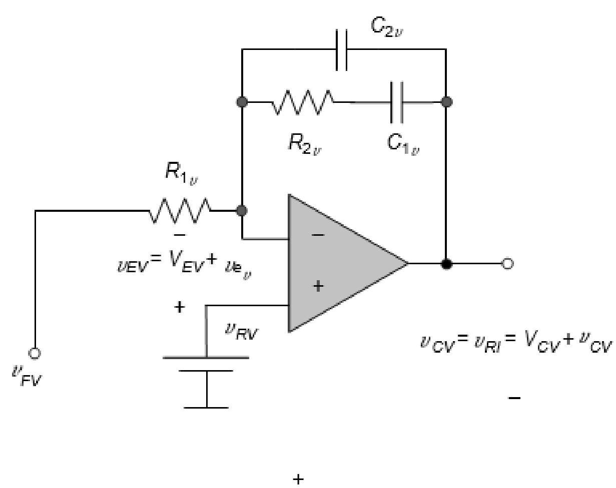


Fig 16 Circuit of Type-II controller used in outer-voltage loop.

giving the geometric mean between the controller corner frequencies as

$$K = \tan\left(\frac{\phi_m}{2} + 45^\circ\right) = 2.83. \tag{2.139}$$

Assuming R_{1v}= 1 kΩ⁄0.25 W⁄2%, the value of C_{2v} is

$$C_{2v} = \frac{|T_v(j\omega_c)|}{2\pi f_c K R_{1v}} = 6.2\ \text{nF} \tag{2.140}$$

A 6.2 nF⁄25 V standard capacitor is selected. The expression for C_{1v} is

$$C_{1v} = C_{2v}(K^2 - 1) = 42 \text{ nF}$$= 42 nF.

A 43 nF∕50 V standard capacitor is selected. The expression for $R_{2v}$is

$$R_{2v} = \frac{K}{2\pi f_c C_{1v}} = 5.2 \text{ k}$$

A 5.2 kΩ∕0.5 W∕5% standard resistor is selected. The transfer function $T_{cv}$of the voltage-loop control circuit is

$$T_{cv}(s) = \frac{v_c(s)}{v_e(s)} = T_{cv0}\frac{1+\frac{s}{\omega_{zcv}}}{s\left(1+\frac{s}{\omega_{pcv}}\right)}$$

where

$$T_{cv0} = \frac{1}{R_{1v}(C_{1v}+C_{2v})} = 20.7 \text{ kV/V}$$

$$\omega_{zcv} = \frac{1}{R_{2v}C_{1v}} = 4.4 \text{ krad/s}$$

$$\omega_{pcv} = \frac{C_{1v}+C_{2v}}{R_{2v}C_{1v}C_{2v}} = 36 \text{ krad/s}$$

Figure 17 shows the theoretically obtained magnitude and phase plots of the designed Type-II controller transfer function used in the outer-voltage loop. The control transfer function $T_{cv}$is incorporated into the critical path to determine the outer-voltage loop gain .

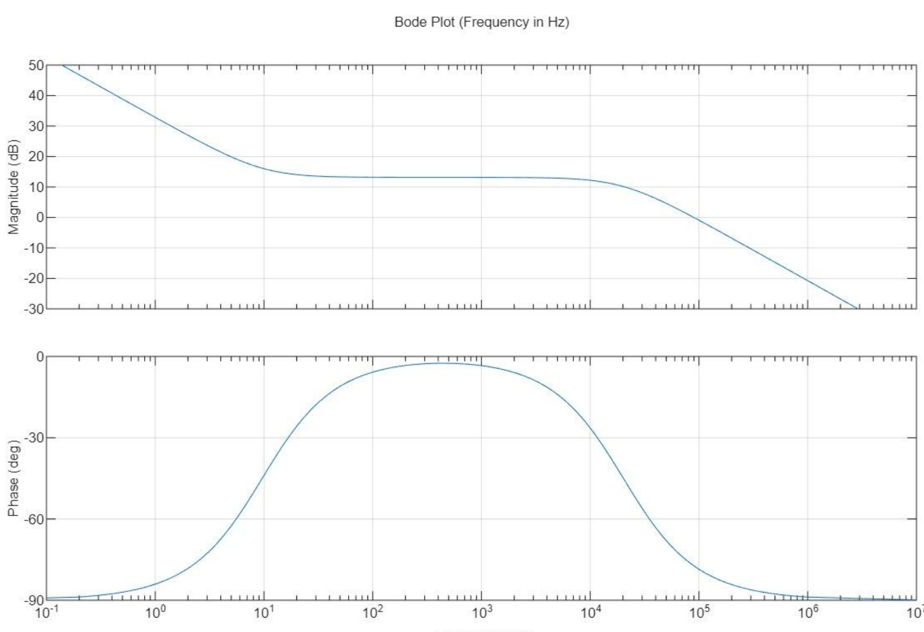
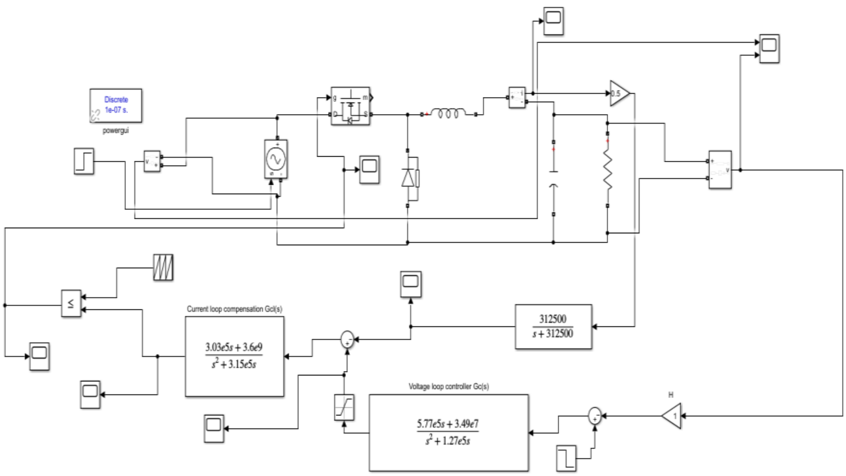


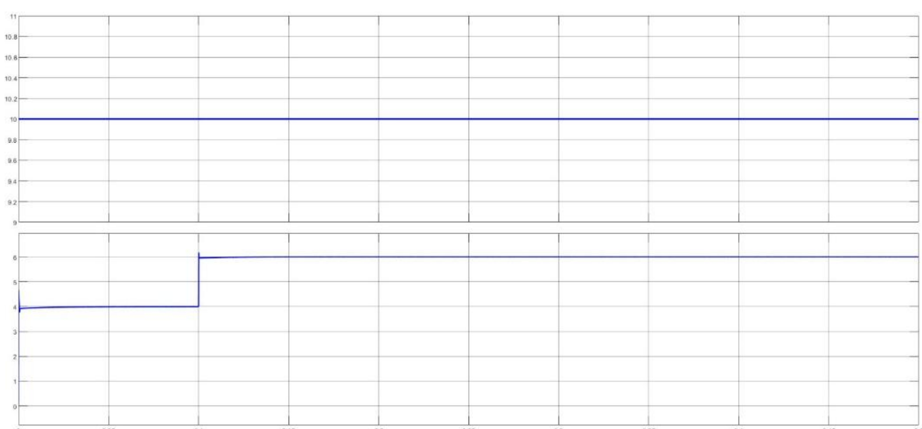
Figure 17 plots of controller transfer function $T_{cv}$of the outer-voltage loop.

Circuit Diagram

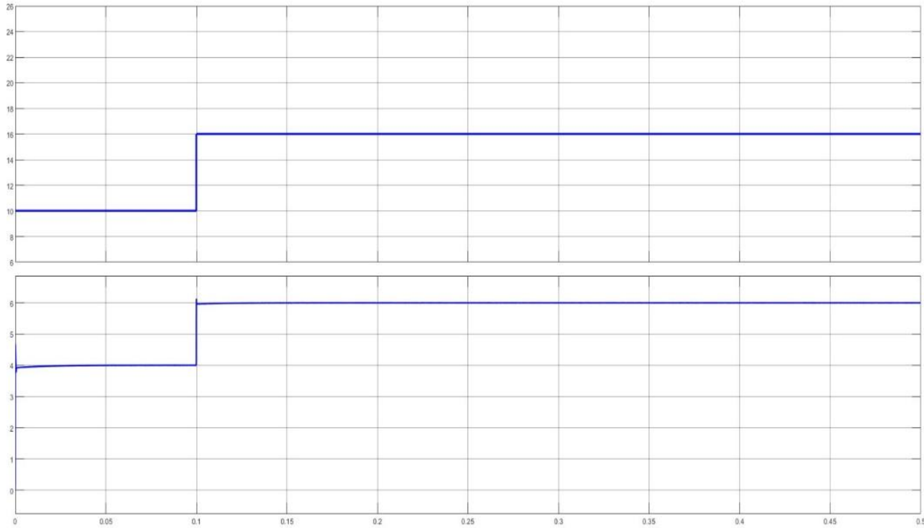


Results :

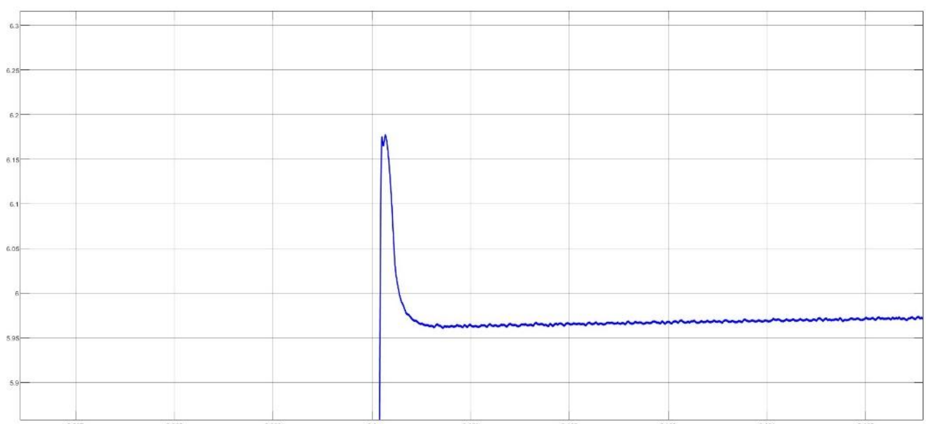
- We change the reference voltage at 0.1 sec.



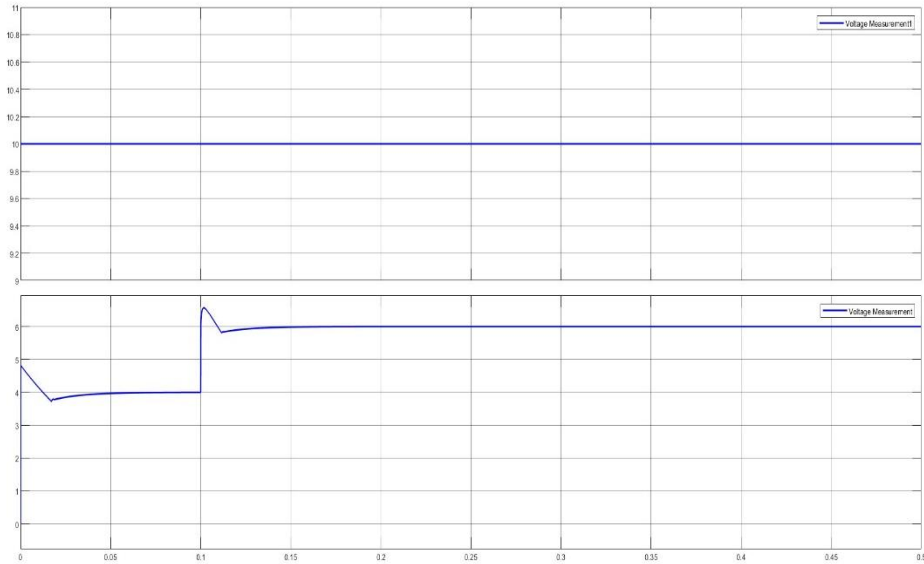
➢ **We change both reference and input voltage at 0.1 sec.**



➢ **Overshot at 0.1sec**



- **At large value of load resistance keeping input voltage same ripple will increase but this can be mitigate by increasing the filter capacitance.**



Conclusion:-

This project analyzed a buck converter controlled using the average current-mode approach under continuous conduction mode. By applying circuit averaging and linearization techniques, the system behavior was studied in both steady-state and dynamic conditions.

The results show that the converter is capable of maintaining the required output voltage with stable operation. The dynamic characteristics of the system are governed by the interaction of the inductor, capacitor, and load, which form a second-order response. The presence of control loops improves system performance by reducing sensitivity to disturbances and enhancing regulation.

The inner current loop enables fast tracking of inductor current and improves the response speed of the converter. The outer voltage loop ensures that the output voltage remains regulated even when there are changes in input voltage or load conditions. The use of compensators helps in achieving sufficient gain and stability margins.

Overall, the control strategy provides reliable performance with good stability and fast response. This makes the converter suitable for practical applications where efficient and stable DC power conversion is required.

References

- **Power Electronics textbooks**

- **Power conversion and control of wind energy systems textbook**

- **https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7793204**

- **Wiley Online Library, "Average Current-Mode Control of Buck DC–DC Converter," [Online]. Available: https://onlinelibrary.wiley.com**

- N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed. Hoboken, NJ, USA: Wiley, 2003.